FINAL REPORT

Team ID: LTVIP2026TMIDS80392

Title: Heart Disease Analysis

1.INTRODUCTION:

1.1 Project Overview

Heart disease remains one of the leading causes of mortality worldwide. Its prevalence continues to rise due to lifestyle changes, unhealthy dietary habits, physical inactivity, and increasing stress levels. Despite advancements in medical science, prevention and early detection remain critical in reducing severe outcomes and improving patient survival rates. Analysing large-scale healthcare data—including patient demographics, medical history, lifestyle choices, and clinical indicators—requires advanced data visualization tools to extract meaningful insights.

1.2 Purpose

The purpose of this project is to analyse heart disease data using Tableau to identify major risk factors affecting cardiovascular health. The project aims to transform raw healthcare data into meaningful and interactive visualizations. It focuses on understanding how age, BMI, lifestyle habits, and medical conditions influence heart disease occurrence. By uncovering patterns and correlations, the analysis supports data-driven decision-making. Ultimately, the project promotes awareness and preventive strategies to reduce heart disease risk.

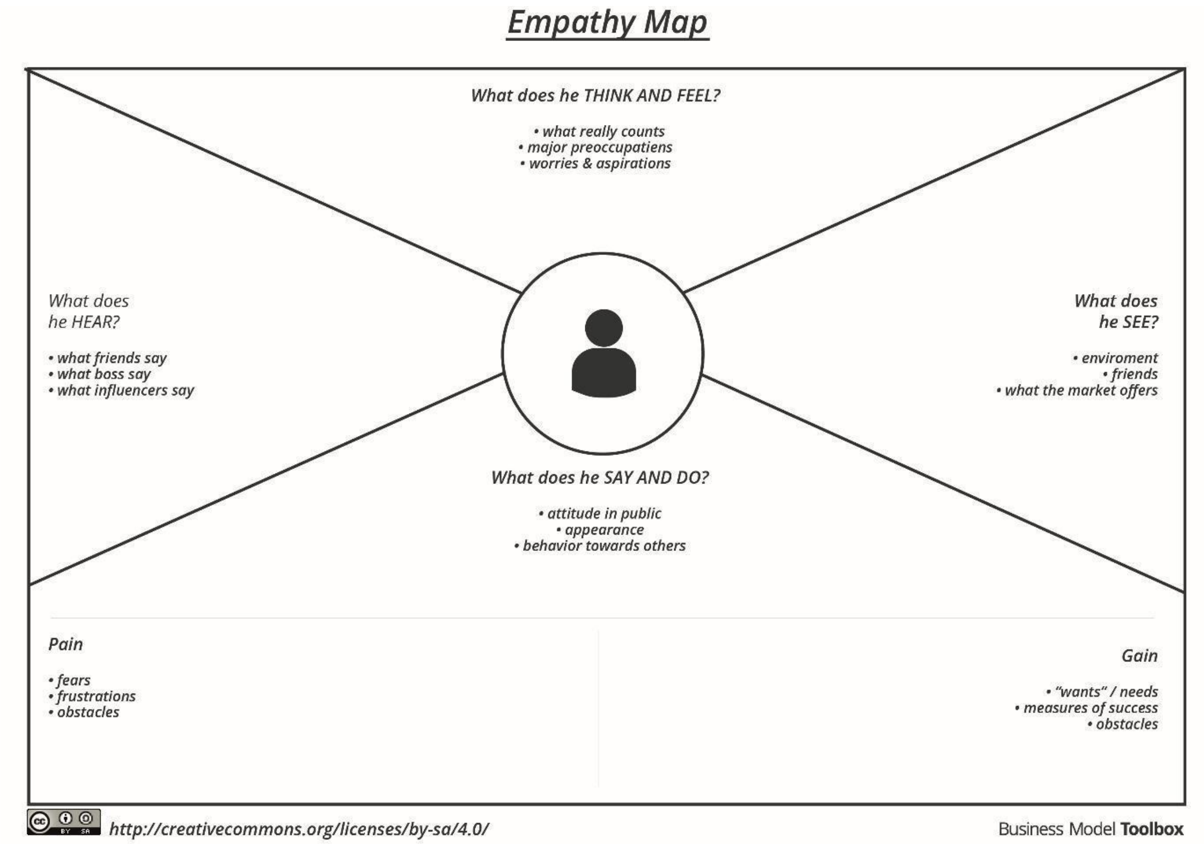
2. IDEATION PHASE

2.1 Problem Statement

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS – 1	A doctor or healthcare professional handling heart disease patients.	Analyze patient health data quickly to identify heart disease risks early.	Manual analysis of multiple medical parameters takes time and may lead to error	Large volumes of patient data make it difficult to identify important risk factors accurately.	Overwhelmed and concerned about missing early warning signs in patients.
PS – 2	A patient	Understand	Medical	There is no	Confused and

	concerned about my heart health.	my risk level and take preventive action early	reports are complex and difficult for me to interpret.	simple system that clearly explains heart disease risk in an easy way.	anxious about my health condition.

2.2 Empathy Map Canvas



2.3 Brainstorming

S. No	Ideas Generated	Category
1	Visualize heart disease distribution across different age groups	Age-based Analysis
2	Compare heart disease occurrence between male and female patients	Gender Comparison
3	Analyze BMI distribution and its relationship with heart disease	Health Risk Assessment
4	Study the impact of smoking and alcohol consumption on heart disease	Lifestyle Analysis
5	Examine the relationship between diabetes and heart disease	Medical Condition Analysis
6	Analyze stroke history and its influence on heart disease occurrence	Risk Factor Evaluation
7	Compare heart disease rates among physically active vs inactive individuals	Behavioral Analysis
8	Highlight states with highest and lowest per capita electricity usage	Health Status Comparison
9	Show changes in urban vs rural electricity consumption	Visualization Enhancements
10	Integrate external datasets like population or weather for deeper insights	Insight & Decision Support

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

Customer Journey Stages:

Stage 1: Problem Awareness

User Realization:

- Heart disease cases are rising due to lifestyle factors.
- Lack of early detection tools.
- Difficulty in comparing patient risk profiles.

Pain Points:

- Raw health data is hard to interpret.
- No centralized dashboard for risk analysis.
- Limited visibility into regional and demographic trends.

Stage 2: Data Exploration

User Actions:

- Access Tableau dashboard
- Filter by age, gender, region, and lifestyle indicators.
- Compare patient groups and risk factors.

System Response:

- Displays interactive graphs and heatmaps
- Highlights top risk contributors (e.g., smoking, high BMI)
- Enables regional and demographic comparisons.

Stage 3: Insight Discovery

User Discovers:

- Middle-aged patients with high cholesterol and smoking habits are at greatest risk
- Urban regions show higher prevalence due to sedentary lifestyle
- Lockdown periods reduced physical activity, increasing risk.
- Certain demographics show consistent upward trends in heart disease.

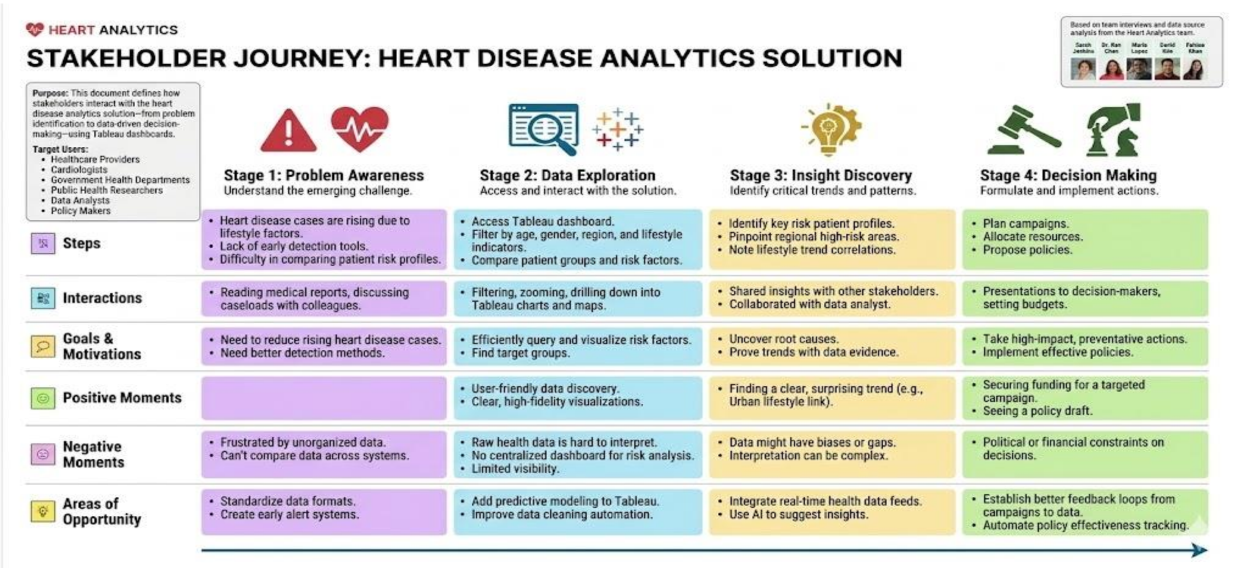
Stage 4: Decision Making

User Decisions:

- Launched targeted awareness campaigns for high-risk groups
- Promote lifestyle changes like physical activity and healthy diets
- Allocate resources for early screening in vulnerable regions
- Develop preventive health policies and incentives

Customer Journey Visualization Flow

- Problem + Data Access + Filtering → Visualization → Insights → Strategic Decision



3.2 Solution Requirement

Functional Requirements:

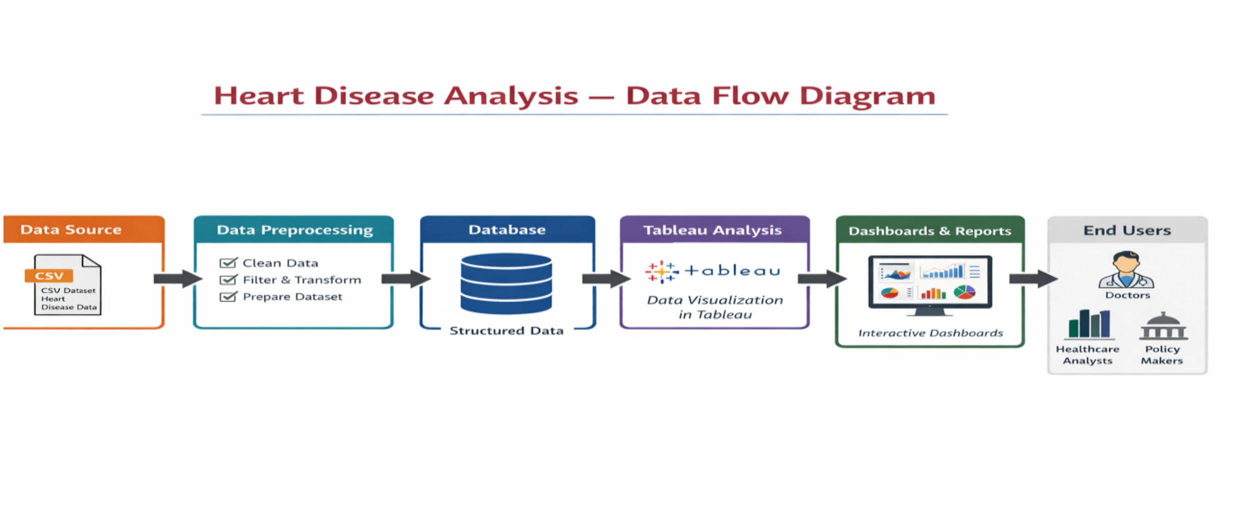
FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form, Registration through Gmail, Registration through LinkedIn
FR-2	User Confirmation	Confirmation via Email, Confirmation via OTP
FR-3	User Login & Authentication	Login using Email & Password, Forgot Password, Session Management
FR-4	Data Upload & Management	Upload patient health data, Validate input data, Store data securely
FR-5	Data Preprocessing	Handle missing values, Feature scaling, Data normalization

Non-functional Requirements:

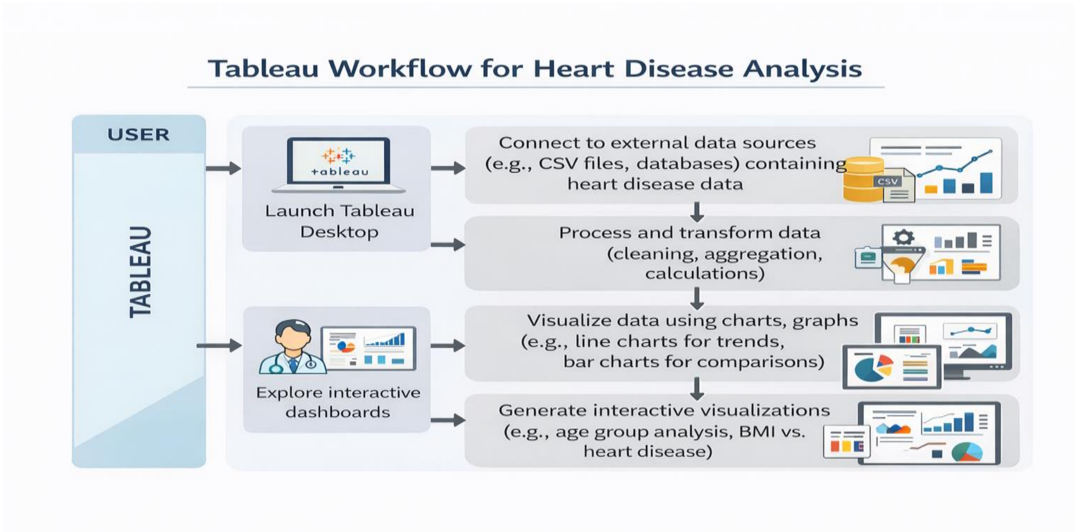
Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should be easy to use with a simple and intuitive interface for users and doctors.
NFR-2	Security	User data and medical information must be securely stored with authentication and authorization mechanisms.
NFR-3	Reliability	The system should provide accurate and consistent prediction results with minimal failures.
NFR-4	Performance	The system should process data and display predictions quickly with minimal response time.
NFR-5	Availability	The application should be accessible whenever required with minimal downtime.
NFR-6	Scalability	The system should support increased users and larger datasets without performance degradation.

3.3 Data Flow Diagram

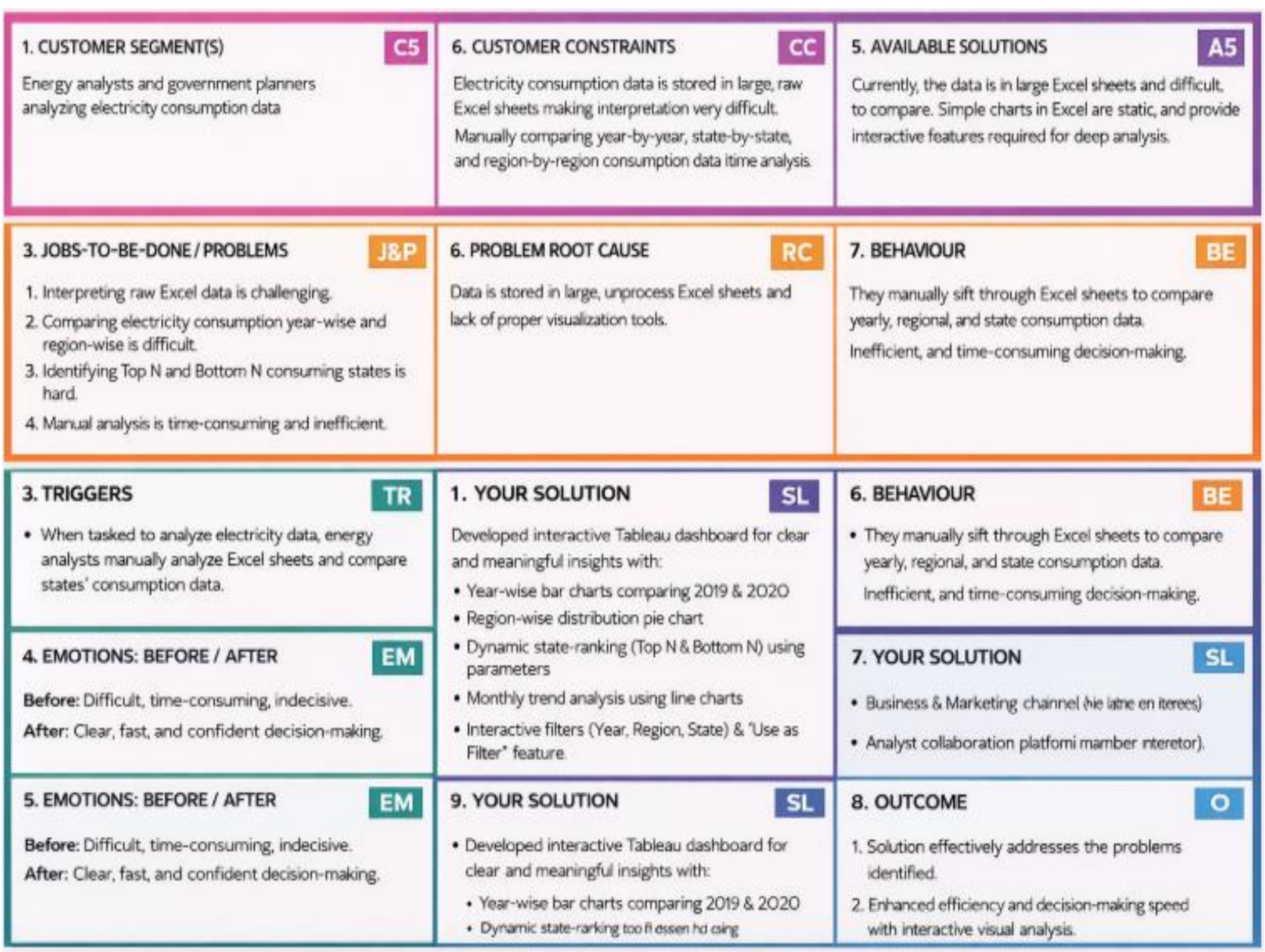


3.4 Technology Stack



4. PROJECT DESIGN

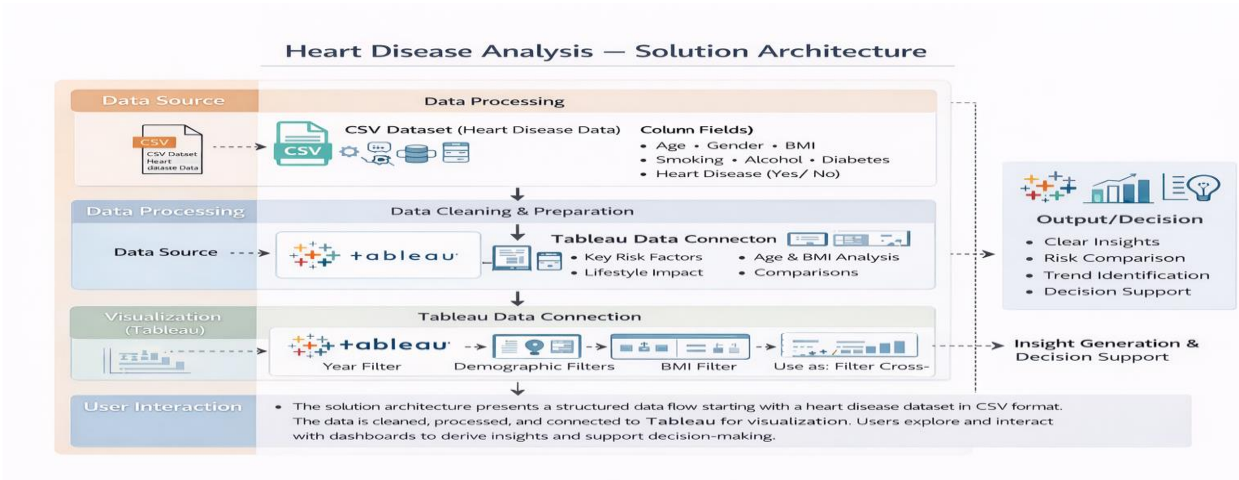
4.1 Problem Solution Fit



4.2 Proposed Solution

S .No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Electricity consumption data is stored in large Excel sheets, making it difficult to interpret, compare year-wise and region-wise trends, and identify high and low consuming states. Manual analysis is time-consuming and inefficient for decision-making.
2.	Idea / Solution description	Develop an interactive Tableau dashboard that visualizes electricity consumption data using bar charts, line charts, and pie charts. The dashboard includes year-wise comparison, region-wise distribution, Top N & Bottom N state analysis, monthly trend visualization, and interactive filters (Year, Region, State).
3.	Novelty / Uniqueness	The solution integrates dynamic filtering, parameter-based Top N and Bottom N analysis, and cross-sheet interaction in a single dashboard. It transforms static Excel data into an interactive, user-friendly analytical system.
4.	Social Impact / Customer Satisfaction	The dashboard helps energy planners and analysts make faster and more accurate decisions. It improves transparency in electricity consumption patterns and supports better energy planning and resource allocation.
5.	Business Model (Revenue Model)	The solution can be adopted by government agencies, electricity boards, and energy companies. It can be implemented as a data analytics service or integrated into energy management systems for performance monitoring.
6.	Scalability of the Solution	The dashboard can be expanded to include additional years, renewable energy data, predictive analytics, and real-time data integration. It can also be adapted for other sectors such as water consumption or fuel usage analysis. predictive analytics, and real-time data integration. It can also be adapted for other sectors such as water consumption or fuel usage analysis.

4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	**Data Collection & Preparation**	USN-1	As an analyst, I can import electricity consumption data into Tableau.	3	High	Shaik Muskan
Sprint-1	**Data Cleaning**	USN-2	As an analyst, I can clean and structure the dataset (Year, Month, Region, State, Usage).	3	High	G.Akshaya
Sprint-2	**Dashboard Creation**	USN-3	As a user, I can view year-wise electricity consumption using bar charts.	5	High	Shaik Muskan
Sprint-2	**Trend Analysis**	USN-4	As a user, I can analyze monthly trends using line charts.	3	Medium	Manjula Praveen Kumar
Sprint-3	**Filter Integration**	USN-5	As a user, I can filter data by Year, Region, and State.	3	High	Dega Venkatarao
Sprint-3	**Top/Bottom Analysis**	USN-6	As a user, I can view Top N and Bottom N states based on usage.	3	Medium	Charan Goud

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-4	**Story & Final Presentation**	USN-7	As a user, I can view insights in story format for better understanding.	4	High	Shaik Muskan

Project Tracker, Velocity & Burndown Chart

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	6	6 Days	15 Jan 2026	20 Jan 2026	6	20 Jan 2026
Sprint-2	8	6 Days	21 Jan 2026	28 Jan 2026	8	28 Jan 2026
Sprint-3	6	6 Days	29 Jan 2026	3 Feb 2026	6	3 Feb 2026
Sprint-4	4	6 Days	4 Feb 2026	7 Feb 2026	4	7 Feb 2026

Velocity:

Total Story Points Completed = 24 Number of Sprints = 4

Average Velocity = 24 ÷ 4 = 6 Story Points per Sprint

If sprint duration = 6 days

Velocity per day = 6 ÷ 6 = 1 Story Point per Day

Burndown Chart:

A Burndown Chart was used to track the remaining work against time during each sprint. The chart shows the ideal work completion line and the actual progress line. The project maintained steady progress, and story points were completed within the planned sprint duration, indicating effective sprint planning and execution.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

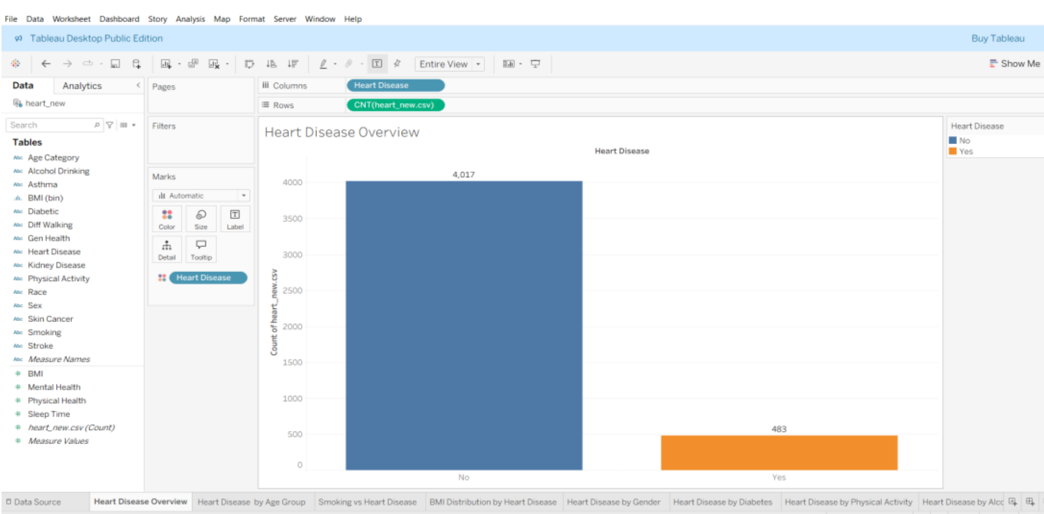
S.No.	Parameter	Screenshot / Values
1.	Data Rendered	The dashboard successfully renders heart disease data without noticeable delay. The dataset includes Age Category, Gender, BMI, Smoking, Alcohol, Diabetes, Stroke, and Heart Disease fields. All visualizations load within acceptable response time.
2.	Data Preprocessing	Data was cleaned before visualization. Categorical fields were verified, BMI bins were created for histogram analysis, and null values were checked and handled appropriately.

3.	Utilization of Filters	Interactive filters such as Age Category, Gender, BMI (bin), and Health Conditions were implemented. “Use as Filter” functionality enables cross-sheet interaction. Dashboards update dynamically when filters are applied.
4.	Calculation fields Used	Calculated fields such as COUNT of patients, BMI bin grouping, percentage calculation, and category-based aggregation were implemented to enhance analysis.
5.	Dashboard design	Number of Visualizations / Graphs – 10 (Bar Charts, Histogram, Comparative Analysis, Percentage Distribution, KPI Overview). Layout is structured, clean, and responsive.
6	Story Design	Story points are arranged logically to explain heart disease trends, risk factors, comparisons, and insights step by step. The storytelling approach improves understanding of the analysis.

7. RESULTS

7.1 Output Screenshots

Dashboard Screenshots:



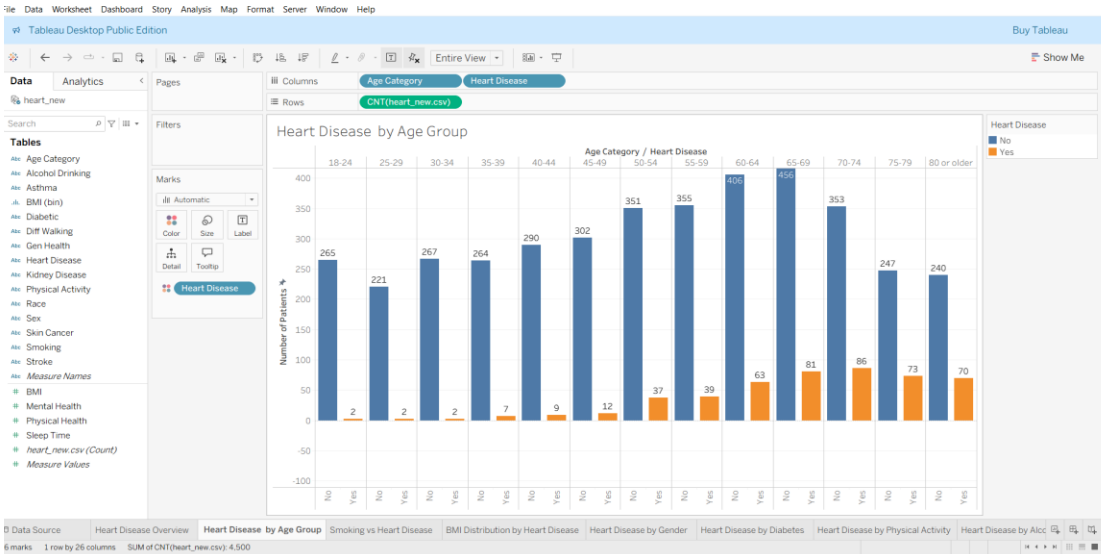


Tableau Desktop Public Edition
Columns: Age Category, Heart Disease
Rows: CNT(heart_new.csv)
Heart Disease by Age Group
Age Category / Heart Disease
18-24 25-29 30-34 35-39 40-44 45-49 50-54 55-59 60-64 65-69 70-74 75-79 80 or older
Number of Patients
265 2 221 2 267 2 264 7 290 9 302 12 351 37 355 39 406 63 456 81 353 86 247 73 240 70
Heart Disease
No
Yes

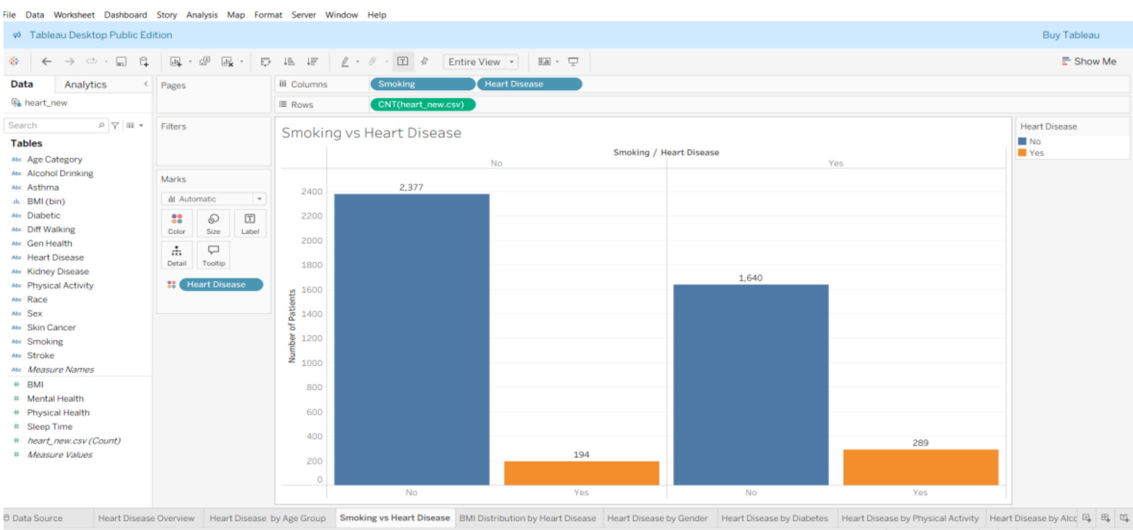


Tableau Desktop Public Edition
Columns: Smoking, Heart Disease
Rows: CNT(heart_new.csv)
Smoking vs Heart Disease
Smoking / Heart Disease
No
Yes
Number of Patients
2,377
194
1,640
289
Heart Disease
No
Yes

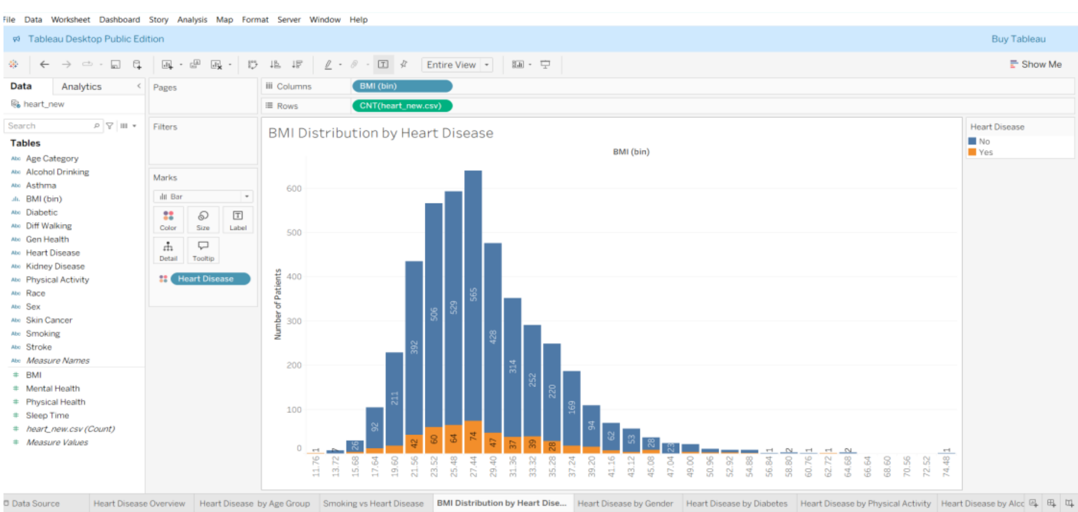


Tableau Desktop Public Edition
Columns: BMI (bin)
Rows: CNT(heart_new.csv)
BMI Distribution by Heart Disease
BMI (bin)
Number of Patients
Heart Disease
No
Yes

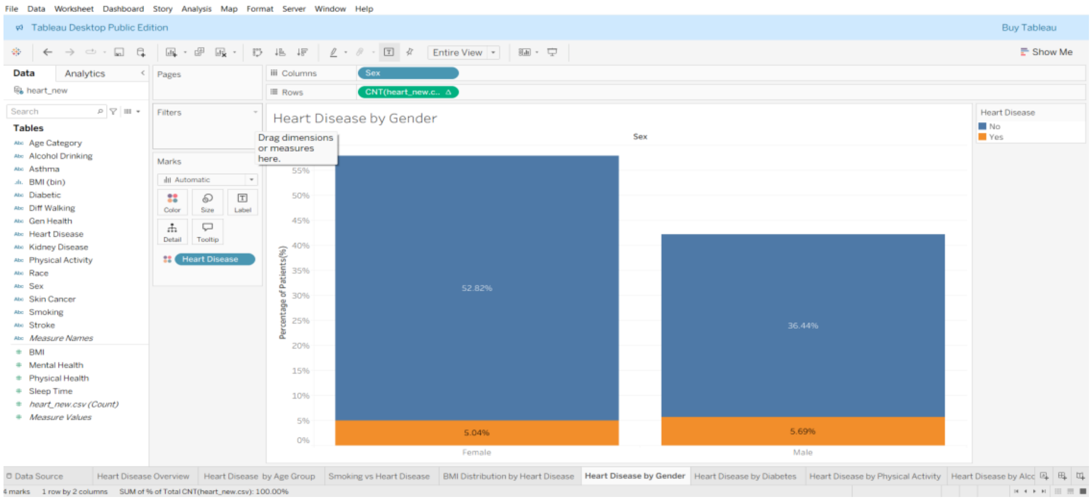


Tableau Desktop Public Edition
Heart Disease by Gender
Sex
Percentage of Patients(%)
52.82%
36.44%
5.04%
5.69%
Female
Male
Heart Disease
No
Yes

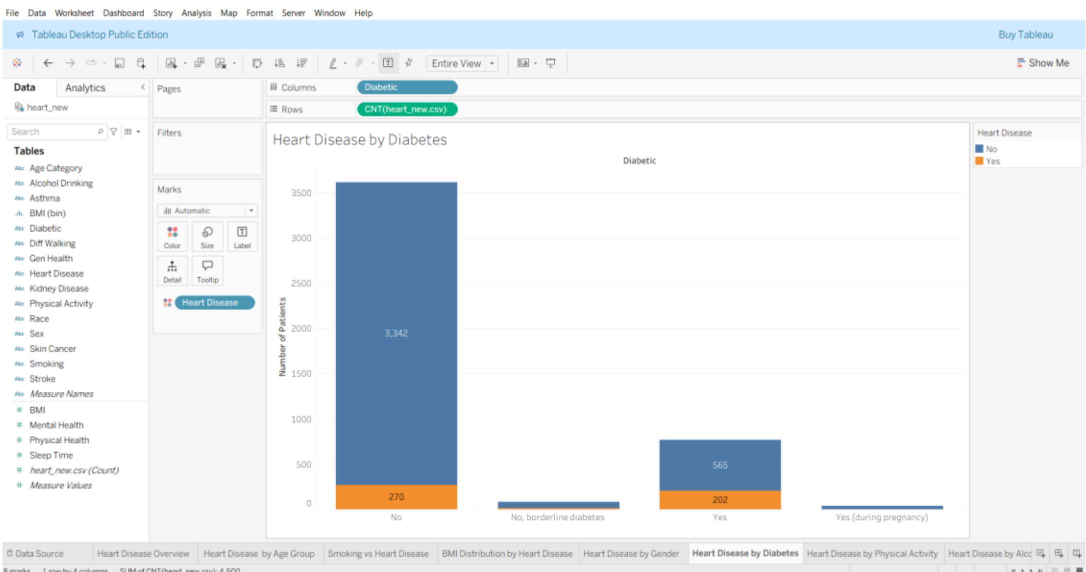


Tableau Desktop Public Edition
Heart Disease by Diabetes
Diabetic
Number of Patients
3,342
270
565
202
No
No, borderline diabetes
Yes
Yes (during pregnancy)
Heart Disease
No
Yes

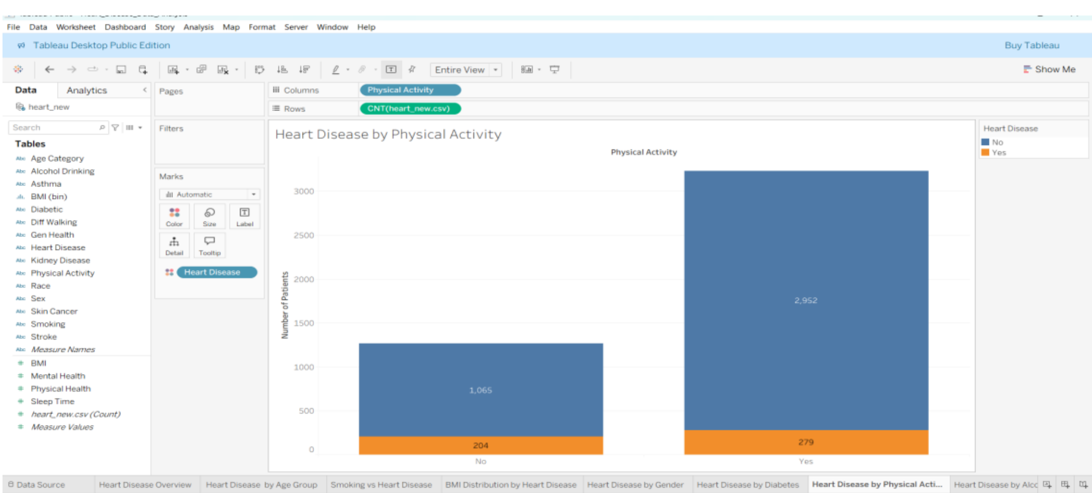
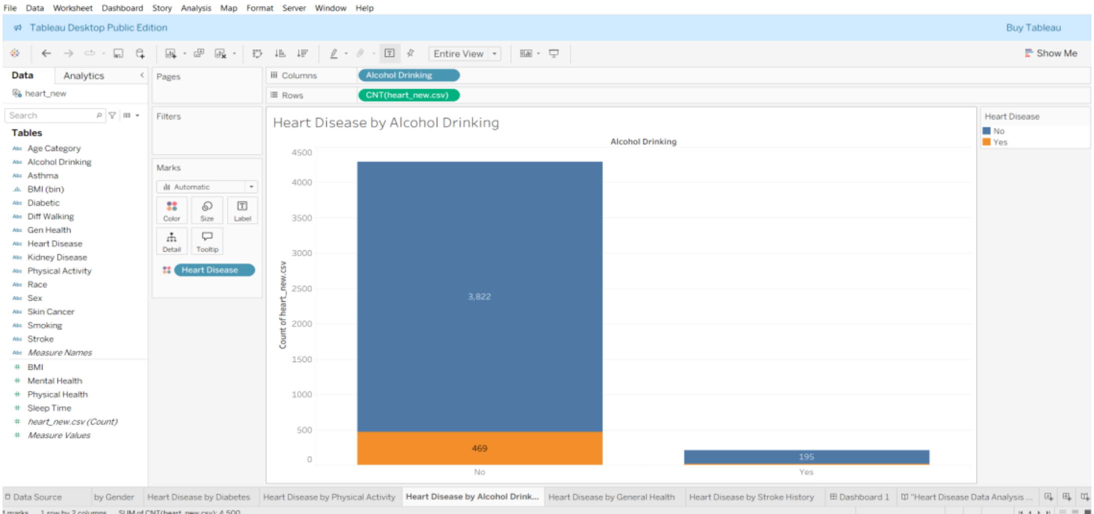
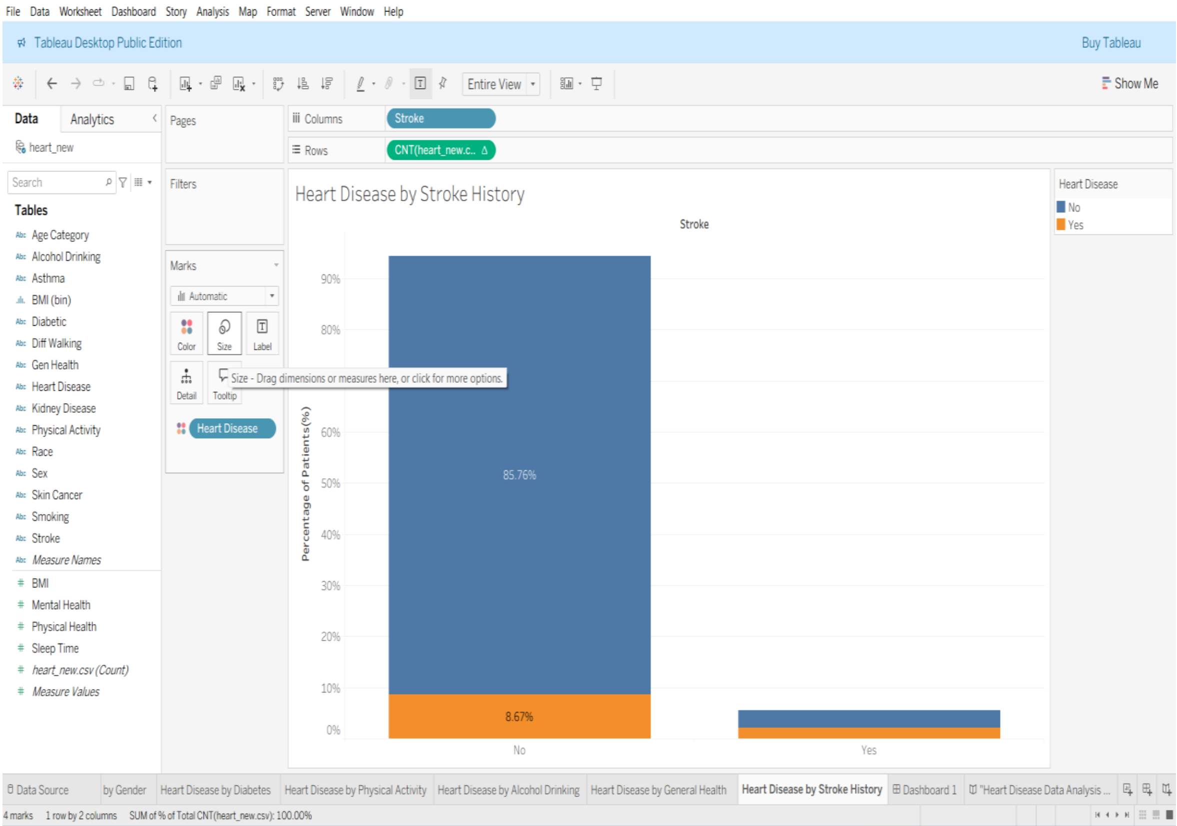


Tableau Desktop Public Edition
Heart Disease by Physical Activity
Physical Activity
Number of Patients
1,065
204
2,952
279
No
Yes
Heart Disease
No
Yes


File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Tableau Desktop Public Edition
Buy Tableau
Entire View
Show Me
Data
Analytics
heart_new
Tables
Age Category
Alcohol Drinking
Asthma
BMI (bin)
Diabetic
Diff Walking
Gen Health
Heart Disease
Kidney Disease
Physical Activity
Race
Sex
Skin Cancer
Smoking
Stroke
Measure Names
BMI
Mental Health
Physical Health
Sleep Time
heart_new.csv (Count)
Measure Values
Pages
Filters
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Heart Disease
Columns
Alcohol Drinking
Rows
CNT(heart_new.csv)
Heart Disease by Alcohol Drinking
Alcohol Drinking
Count of heart_new.csv
4500
4000
3500
3000
2500
2000
1500
1000
500
0
3,822
469
195
No
Yes
Heart Disease
No
Yes
Data Source
by Gender
Heart Disease by Diabetes
Heart Disease by Physical Activity
Heart Disease by Alcohol Drink...
Heart Disease by General Health
Heart Disease by Stroke History
Dashboard 1
"Heart Disease Data Analysis ...


File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Tableau Desktop Public Edition
Buy Tableau
Entire View
Show Me
Data
Analytics
heart_new
Pages
Columns
Stroke
Rows
CNT(heart_new.c.. Δ
Filters
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Size - Drag dimensions or measures here, or click for more options.
Heart Disease
Tables
Age Category
Alcohol Drinking
Asthma
BMI (bin)
Diabetic
Diff Walking
Gen Health
Heart Disease
Kidney Disease
Physical Activity
Race
Sex
Skin Cancer
Smoking
Stroke
Measure Names
BMI
Mental Health
Physical Health
Sleep Time
heart_new.csv (Count)
Measure Values
Heart Disease by Stroke History
Stroke
Percentage of Patients(%)
90%
80%
60%
50%
40%
30%
20%
10%
0%
85.76%
8.67%
No
Yes
Heart Disease
No
Yes
Data Source
by Gender
Heart Disease by Diabetes
Heart Disease by Physical Activity
Heart Disease by Alcohol Drinking
Heart Disease by General Health
Heart Disease by Stroke History
Dashboard 1
"Heart Disease Data Analysis ...
4 marks 1 row by 2 columns SUM of % of Total CNT(heart_new.csv): 100.00%

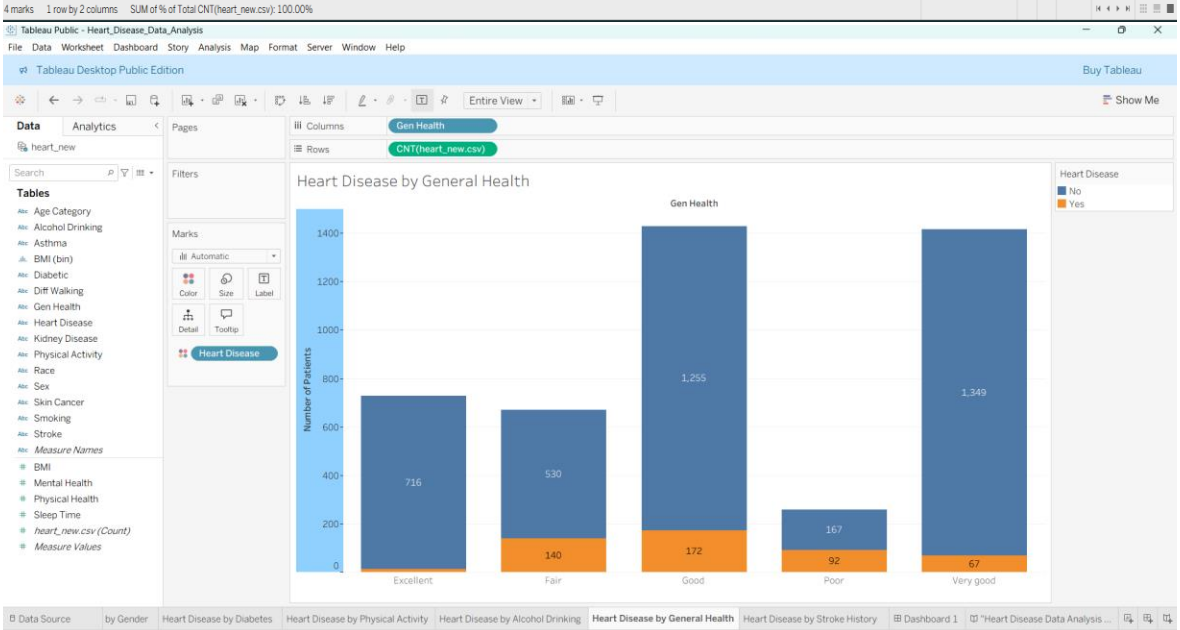


Tableau Public - Heart_Disease_Data_Analysis
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Tableau Desktop Public Edition
Buy Tableau
Entire View
Show Me
Data
Analytics
heart_new
Pages
Columns
Gen Health
Rows
CNT(heart_new.csv)
Filters
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Heart Disease
Tables
Age Category
Alcohol Drinking
Asthma
BMI (bin)
Diabetic
Diff Walking
Gen Health
Heart Disease
Kidney Disease
Physical Activity
Race
Sex
Skin Cancer
Smoking
Stroke
Measure Names
BMI
Mental Health
Physical Health
Sleep Time
heart_new.csv (Count)
Measure Values
Heart Disease by General Health
Gen Health
Number of Patients
1400
1200
1000
800
600
400
200
0
716
530
140
1,255
172
167
92
1,349
67
Excellent
Fair
Good
Poor
Very good
Heart Disease
No
Yes
Data Source
by Gender
Heart Disease by Diabetes
Heart Disease by Physical Activity
Heart Disease by Alcohol Drinking
Heart Disease by General Health
Heart Disease by Stroke History
Dashboard 1
"Heart Disease Data Analysis ...
10 marks 1 row by 5 columns SUM of CNT(heart_new.csv): 4,500

Dash Board

Story:

8. ADVANTAGES & DISADVANTAGES

Advantages

1. **Interactive Visualization**
   The dashboard provides interactive charts and filters that allow users to explore electricity consumption data dynamically.
2. **Improved Decision Making**
   Year-wise and region-wise analysis helps stakeholders make informed and data-driven decisions.
3. **Easy Trend Identification**
   Monthly and seasonal trends can be easily identified through line charts and comparative analysis.
4. **Time Saving**
   Instead of manually analyzing raw Excel data, the dashboard presents insights instantly.
5. **User-Friendly Interface**
   The system is simple to use and does not require advanced technical knowledge.
6. **Better Resource Planning**
   Helps energy planners identify high and low consumption regions for efficient energy distribution.

Disadvantages

- **Dependent on Data Quality**

If the input data contains errors or missing values, the analysis may not be accurate.

- **Limited to Available Data**

The dashboard can only analyze the data provided and cannot predict future consumption without advanced modeling.

- **Requires Tableau Software**

Users need access to Tableau to interact with the dashboard.

- **Initial Setup Time**

Data cleaning and dashboard development require initial time and effort.

9. CONCLUSION

This project analysed heart disease data using Tableau to identify key risk factors and trends. Interactive dashboards were created to study the impact of age, BMI, lifestyle habits, and medical conditions. The analysis revealed significant relationships between heart disease and factors such as diabetes, stroke, and physical inactivity. Data visualization helped transform raw healthcare data into meaningful insights. Overall, the project demonstrates the importance of data-driven decision-making in preventive healthcare.

10. FUTURE SCOPE

The project can be enhanced by integrating real-time healthcare datasets for continuous analysis. Machine learning models can be applied to predict heart disease risk more accurately. The dashboard can be deployed as a web application for wider accessibility. Advanced analytics such as trend forecasting can be incorporated. Additional health indicators can be included to improve risk assessment and decision support.

Dataset Link

https://drive.google.com/file/d/190Qmq27LeZZ_nWricP3Obl7ys_5otEsp/view

Project Demo Link

https://drive.google.com/file/d/1w5htiXVzehzcuh-LtqUM3ilTFsr9Mez1/view?usp=sharing

Git hub Link

https://github.com/Bhavana137/heart-disease-tableau-analysis